

AKQ: A hybrid quantum-classical image encryption system

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Abstract—Image encryption is an essential field between computer vision and security which aims to protect the image before future attacks. Nowadays, it is threatened by quantum computers, which can break some classical cryptography. The encryption system has been upgraded from quantum key exchange protocol to post-quantum key-establishment mechanisms, but they are not explicitly designed for the image encryption task. In this paper, we propose a hybrid quantum-classical theoretical framework and its instance AKQ system which is more secure and suitable in the post-quantum era.

Index Terms—hybrid quantum-classical, image encryption, post-quantum cryptography, security

I. INTRODUCTION

In recent years, the use of images has become ubiquitous in various applications, especially critical in fields such as social networks, medical and military applications, where the confidentiality and integrity of images play an important role. In particular, with the increasing dependence on the internet and other digital communication channels, securing the transmission and storage of images has become essential to prevent unauthorized access. Moreover, the use of high-resolution images and videos and the increasing use of cloud storage have made it easier for attackers to access and manipulate image data.

Numerous efficient schemes have been introduced and implemented across various domains. However, the majority of them rely on classical asymmetric cryptography, such as Rivest–Shamir–Adleman (RSA) [1], which is forecast to be attacked by quantum computers. Quantum computers leverage algorithms such as Shor’s [2] and Grover’s algorithms [3] to exploit vulnerabilities in these classical cryptographic systems. The development of quantum computers has seen exponential progress, exemplified by the transition from IBM Falcon’s 27-qubit configuration in 2019 to IBM Osprey’s

433-qubit configuration in 2022, with an anticipated advancement to approximately 4000 qubits by 2025. Although quantum computers need tens or hundreds of years from now to be able to attack classical cryptography, we need to use a higher secure image encryption system to protect data from the past. As a result, there is a growing demand for new image encryption techniques that can provide robust protection against various types of attacks, including classical attacks and quantum attacks.

II. RELATED WORKS

Image encryption relates closely to the image exchange process, as described in Fig. 1. There are three sides in an image exchange scene: sender, receiver, and attacker. In the symmetric encrypt and decrypt processes, the sender encodes an image with a secret key k , the receiver must use the same key to decode and receive the original image. The attacker can either attack the unsecured channel/database and key establishment system to steal users’ images. Differ from other datatypes, the patterns will not be hidden if we use a standard1 encrypt/decrypt process, as *cimg* in Fig. 1, and the attacker can get valuable information if we don’t delete it. Therefore, researchers have proposed various image encryption techniques, which focus almost on spatial domain-based image encryption techniques, as below.

Chaos-based encryption techniques [4, 5] is another popular approach to encrypting images. These techniques utilize chaotic maps to scramble the pixel values in a manner that is unpredictable and difficult to reverse. Chaos-based encryption algorithms are highly effective in protecting image data from unauthorized access.

Recent research has explored the use of **neural networks** [6, 7], these techniques utilize deep learning

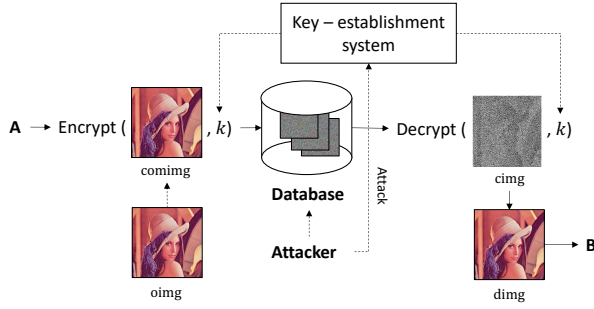


Fig. 1. Overview about the image exchange process. **A**, **B**, k , $oimg$, $cimg$, and $dimg$ denote sender, receiver, secret key, original image, cipher image, and decrypted image, respectively. Classical image encryption systems have two components: key-establishment and encrypt/decrypt components.

models to learn the encryption process from the pairs $\{oimg, cimg\}$ generally. When the trained neural networks are only shared between sender and receiver, the $cimg$ will be hard to decrypt. However, we take a cost for the training and testing process.

Steganography-based encryption techniques [8, 9, 10] involve hiding encrypted image data within other images or media files that human eyes can not see. These techniques aim to provide an additional layer of security by concealing the encrypted data in plain sight. However, this approach can be vulnerable to attacks targeting the steganography algorithm.

DNA-based encryption techniques [11, 12] have recently emerged as a novel approach to image encryption. These techniques utilize DNA sequences to encrypt image data, which can be decoded using DNA sequencing technology. While this approach is still in the early stages of development, it has shown promising results in terms of security and data storage capacity.

These above image encryption techniques offer security, efficiency, and robustness before classical attacks such as differential analysis and semantic analysis. In the quantum era, these techniques can be broken by quantum brute-force attack because its keyspace increases by key size slower than exponential [13] of the number of qubits. The pseudo-random module does not adequately randomize the distribution of pixel values, an attacker might perform frequency analysis on the $cimg$ to gain insights into the $oimg$. Therefore, the new image encryption system should recover those two weaknesses: the key space and the true random number generator.

III. APPROACH

A. Hybrid quantum-classical system

Quantum cryptography offers the absolute secure protocol before the quantum attack by using the quantum

mechanism itself but the deployment cost prevents us from applying it into real production. Taking inspiration from quantum machine learning [14] and the hybrid quantum-classical computation physics model [15], we propose a hybrid approach that combines classical and quantum components and takes advantage of both the quantum and classical approaches. In this system, classical components conduct almost all calculations whereas quantum computing is used to execute some hard tasks such as random number generator (RNG) or compressing high-resolution images.

The structure of the classical part looks like current image encryption techniques where it mainly has a key establishment and encrypt/decrypt components:

(1) The encrypt component divides data into blocks of fixed sizes, and each block is encrypted independently by k . The decrypt component is the inverse version of the encrypt component which uses the same k .

(2) The key-establishment system helps A and B share the same k via asymmetric mechanism, where A encodes k to c by B 's public key, and B decode it by his private key. In this component, c is known as the intermediate parameter.

(3) The compression components decrease the amount of data and delete the patterns on the image, it can be integrated with the encrypt component which has diffusion property, or operates independently depending on the system design.

(4) The quantum part can compose useful components including RNG, compressed image, and quantum key exchange (QKE).

B. AKQ system

By applying the above scheme in the real application, we propose a new image encryption system, including AES encrypt/decrypt component (A), CRYSTALS-Kyber (K), Quantum RNG (Q_1), and Quanvolutional filter component (Q_2):

(A) We choose AES [16] as the encrypt/decrypt component because it is a widely used symmetric-key encryption algorithm that has been extensively studied and adopted in various fields, including image encryption. To use AES, the data is first divided into blocks of fixed sizes, and each block is encrypted independently (ECB mode [17]) or processed with feedback from the previous block (in modes such as CBC, CFB, or OFB), depending on the chosen mode of operation. AES is considered one of the most secure encryption algorithms available today, with a high level of resistance to attacks such as brute-force and differential attacks, specifically, the AES-256 version which makes it difficult for the future quantum computer to break the cipher image. Additionally, AES is computationally efficient, which makes it suitable for use in the application aspect.

However, there are also some limitations to using AES for image encryption. One of the main challenges is that AES encryption can be vulnerable to some attacks, such as side-channel attacks, which can exploit weaknesses in the implementation of the algorithm. Since the file size is huge, it can be suspected to contain some critical information. We recommend ECB mode because of the fastest speed in modes. However, ECB does not provide any form of data chaining or diffusion, and can not hide the patterns on cipher images, commonly, this weakness is fixed by applying a chaotic map before using AES. E.g: Arnold's cat map [18], hyper-logistic map [19], exponential map [20], Lorenz attractor [21], ... In our system, this task is responsible for component (Q_2).

(K) CRYSTALS-Kyber [22, 23, 24] is a post-quantum key exchange algorithm designed to provide secure communication in a world where quantum computers break classical cryptographic schemes. It is based on the problem of learning with errors (LWE), which is believed to be hard for quantum computers to solve efficiently. This algorithm offers several security levels, depending on the chosen parameters, and has been designed to be fast and efficient in both software and hardware implementations. It has been extensively tested and evaluated by the cryptographic community and is considered one of the leading candidates for post-quantum cryptography. This algorithm aims to identify and standardize PQC schemes for future use.

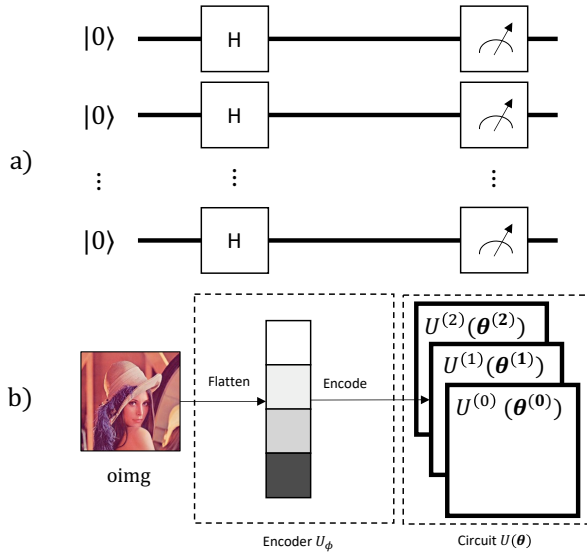


Fig. 2. a) The n -qubit Quantum RNG circuit is made by n Hadamard gates b) A scheme of quanvolutional filter, which is based on QNN variants in Ref. [25]. The quanvolutional filter processes patch by patch until the end of the image. U_ϕ converts each patch into a quantum state by flattening, normalizing, and compressing by a series of parameterized gates such as $\{R_j, CR_j\}$ with $j \in \{x, y, z\}$.

(Q_1) Quantum RNG is needed in the key generation

process to create a random key as well as for other components such as initialization vectors (IV) and nonces. Quantum RNG is known as a True Random Number Generator (TRNG) and provides a source of randomness where all classical algorithms that run on classical computers are Pseudo RNG, RNG must be capable of generating random numbers at least as quickly as they are consumed therefore it not fit with image encryption purpose. To create TRNG, we apply Hadamard gates to qubits and receive state $|\psi\rangle$ that represents a blend of 0 and 1 with equal probabilities, a clear manifestation of quantum superposition. When the qubit is measured, it collapses into one of these classical states probabilistically, providing an inherently random outcome. This randomness arises from the inherent uncertainty and probabilistic nature of quantum mechanics. They generate truly unpredictable sequences of numbers with profound implications for encryption. With H and $\mathcal{M}$ denotes for Hadamard gate and measurement operator, respectively, we have a quantum circuit that runs on n -qubits:

$$|\psi\rangle = H^{\otimes n}|0^{\otimes n}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)^{\otimes n}; \quad (1)$$

$$\bar{a}_j = \mathcal{M}(|\psi\rangle)$$

where probability when receiving value of $\bar{a}_j$ is 0 or 1 is $\frac{1}{2^n}$ if $n = 1$, 00 or 01 or 10 or 11 is $\frac{1}{2^2}$ if $n = 2$ and so on, note that $|\bar{a}_j| = N = 2^n$. We conduct measurements m time and get the binary-array $\bar{a}_0 \bar{a}_1 \dots \bar{a}_m$, after that convert it into z format. The number of measurements will decrease exponentially based on n .

(Q_2) Quanvolutional filter: Using quantum state preparation algorithm (QSP) [26], we can compress any images by ansatz U and its parameter θ with exponential scale where we only use $\log_2 NM$ qubits to process $N \times M$ images.

$$\theta \leftarrow QSP(Nom(Vec(oimg, U)))$$

$$U(\theta) = \bigotimes_{k=0} (U_k(\theta_k) \otimes V_k) \quad (2)$$

where Vec denotes for Vectorization function, $Nom(x) : x \in \mathbb{R}^N \rightarrow \hat{x} = \frac{x}{|x|}$, U_k, V_k denotes for parameterized and fixed quantum gates. The compressed image can be decompressed by $|\dim\rangle = U(\theta)|0^{\otimes n}\rangle$.

The above process looks like DNA-based image encryption techniques where an image is encoded from image form to another format by DNA operations. The difference between DNA-based techniques and Q_2 is that $|\theta|$ is polynomially increased by n which means Q_2 compressed image effectively in the case of high-resolution images. If we do not have enough qubits for the quantum system, we will divide the image into

patches and process patch by patch. In this way, the dimensional of θ changes from $poly(n)$ to $poly(poly(n))$.

$$\begin{aligned} oimg &:= \{\text{patch}^{(0)}, \text{patch}^{(1)}, \dots, \text{patch}^{(k)}\} \\ &= \{U^{(0)}(\theta^{(0)}), U^{(1)}(\theta^{(1)}), \\ &\dots, U^{(k)}(\theta^{(k)})\} \end{aligned} \quad (3)$$

where $\theta := (\theta^{(0)}, \theta^{(1)}, \dots, \theta^{(k)})$. There are different values and sizes of θ with each image and U . We must choose a suitable U and run QSP to find it. This step is conducted on the sender side and also had $poly(n)$ time complexity. There are a trade-off between the $distance(dimg, oimg)$ and $|\theta|$. The larger $|\theta|$ or the larger optimization cost, the smaller $distance(dimg, oimg)$.

Algorithm 1 Scheme for image encryption

- 1: (B) $z \leftarrow \text{QuantumRNG}()$
 - 2: (B) $\theta \leftarrow \text{Compress}(oimg, U)$
 - 3: Initial parameters for **Kyber512** and **AES256**
 - 4: (B) $pk, sk \leftarrow \text{Kyber512}$. GenerateKey(z) [27].
 - 5: (A) $pk \leftarrow \text{ReadFromCASystem}()$
 - 6: (A) $c, k \leftarrow \text{Kyber512}$. Encode(pk)
 - 7: (A) $\theta' \leftarrow \text{AES256}$. Encrypt(θ, k).
 - 8: (A) $cimg \leftarrow \{\theta', U, c\}$
 - 9: (A) Send $cimg$ to (B) through an insecure channel or save it to the database.
 - 10: **return**
-

Algorithm 2 Scheme for image decryption

- 1: Using A's configuration for **Kyber512** and **AES256**.
 - 2: (B) Receive $cimg = \{\theta', U, c\}$.
 - 3: (B) $k \leftarrow \text{Kyber512}$. Decode(c, sk).
 - 4: (B) $\theta \leftarrow \text{AES256}$. Decrypt(θ', k).
 - 5: (B) $dimg \leftarrow \text{Decompress}(\theta, U)$
 - 6: **return**
-

The image encryption system includes the following three kinds of processes: compress/decompress, exchange secret key, and symmetrically encrypt/decrypt, which is presented in the Algorithm. 1, Algorithm. 2 and Fig. 3. In the beginning, A and B agree on the component configurations: secret key length and ansatz U . After that, the compress component deletes all patterns in the original image and compresses $oimg$ as ansatz U and its parameter θ . Next, AES encrypts θ with a secret key which is generated from the encoder. In this system, ECB is applied because of its speed, and the deleting pattern task is conducted by the compress component. The parameter c combines with $B'sk$ which helps to decrypt and decompress the cipher image. Depending

on the requirements of the application, we can adjust the key size and parameters. The bigger the key size, the higher the security, and the more resources the encrypt / decrypt process takes.

IV. EVALUATIONS

A. Key security analysis

The AES-256 which combines with CRYSTALS-Kyber-512 provides a secure level of 5, the highest degree of security against unauthorized access based on The NIST Post-Quantum Crypto "Competition" [28]. In the CRYSTALS-Kyber-512 variant, the public key, secret key, and parameter c consist of 800, 1632, and 768 bytes, respectively [27]. The key space of CRYSTALS-Kyber-512 refers to the total number of possible secret keys that can be generated by the algorithm. Since the secret key size is 1632 bytes, the key space can be calculated as $2^{(8 \times 1632)}$, which is an extremely large number. This immense key space makes exhaustive key search strategy from quantum computers infeasible.

B. Execution time analysis

AES itself does not have inherent scalability, it can be implemented in a scalable manner using appropriate infrastructure and cryptographic techniques. CRYSTALS-Kyber is designed to be efficient and scalable for a wide range of applications, including those with limited computational resources such as low-power devices or IoT devices. Additionally, CRYSTALS-Kyber is designed to be parallelizable, meaning that it can be implemented to run efficiently on systems with multiple processing cores. This allows for better scalability in terms of performance when handling larger volumes of data. Quantum components, in theory, can process large data with exponential scaling, which are suitable for high-resolution images.

C. Weakness

Encoding $oimg$ as $\{\theta', U, c\}$ reduced the amount of data and deleted the patterns but lost information on $dimg$. In detail, Q_2 offers the user the trade-off between the quality of $dimg$ and $|\theta'|$. As results from Ref. [26], the trace distance between an arbitrary and reconstructed state will increase fast by the number of qubits if we use the same structure of U . Therefore, the user must be careful when choosing U depending on their target. QSP also requires the input dimensional as exponential of 2 which means we must add padding to undersized images. Nowadays, quantum computers are not popular and have a limited number of qubits, which is both an advantage and disadvantage for this system. The disadvantage is that the sender must use quantum components via cloud services, which leads

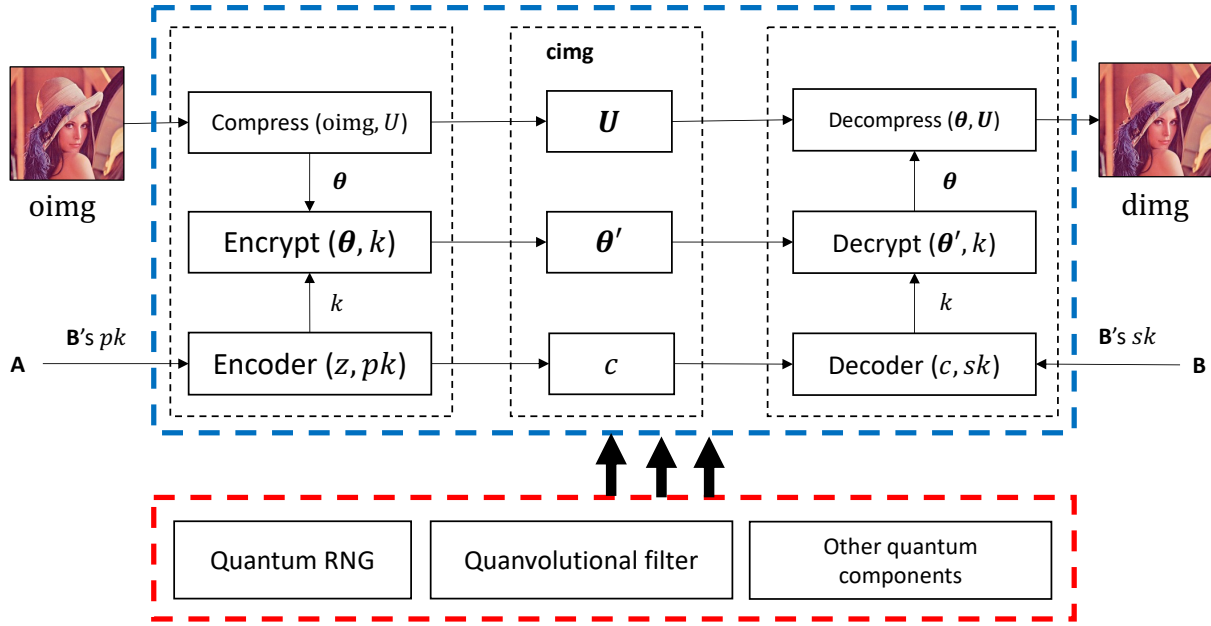


Fig. 3. The diagram of the overall hybrid quantum-classical image encryption schemes. There are classical components and quantum components. While classical components (blue border) operate on both sides, quantum components (red border) only operate on the sender side. Classical and quantum components run on classical and quantum computers, respectively, sender A can use quantum components via a cloud system.

to other attack opportunities such as man-in-the-middle attacks, and API vulnerabilities.

V. CONCLUSION

This paper proposes a new theoretical framework for image encryption which is enhanced from hybrid models in other fields, this system not only refuses both classical and quantum attacks but also runs on normal computers. An instance of this framework is AKQ: which consists of AES, CRYSTALS-Kyber, Quantum RNG, and Quanvolutional filter. AES is highly efficient and scalable for high-performance encryption, CRYSTALS-Kyber is designed to be secure under quantum attack scenes, and two quantum components execute their task at an exponential scale. Quantum RNG helps the system be more secure while the Quanvolutional filter makes *cimg* lighter.

In conclusion, this system is suitable for real-world image encryption applications, especially in the post-quantum era [29] because of the availability of quantum packages such as **qrand**¹ (for Quantum RNG) and **qtm**² (for Quanvolutional filter). To assess AKQ more objectively, metrics include number of pixel change rate (NPCR), correlation coefficient (CC), information entropy, and execution time will be evaluated later.

¹<https://github.com/pedrorrivero/qrand>

²<https://github.com/vutuanhai237/qtm>

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